Problem 2 | Problems T3_Diffusion.pdf #2 | Arrhenius · Semiconductor Doping

Diffusion of Aluminum in Monocrystalline Silicon

Original statement · Problems T3_Diffusion.pdf

2.- Al *can diffuse in a* Si *monocrystal. Calculate the temperature at which the diffusion coefficient will have a value of* $10^{-14}\ \mathrm{m^2/s}$*? Data:* $E_D = 73\ \mathrm{kcal/mol}$*;* $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$ *and* $R = 1.987\ \mathrm{cal/mol \cdot K}$. Official Solution: T = 1566 K (1293 °C)

PHASE 1 · HYPOTHESES & PARAMETERS

- **Volume Diffusion in a Single Crystal:** A single crystal has no grain boundaries, so diffusion takes place through the volume, the slowest path ($D_{surface} > D_{grain\ boundary} > D_{volume}$) [Slide 25].
- **Arrhenius Behavior:** The temperature dependence of D obeys Arrhenius kinetics with constant D_0 and E_D [Slide 26].
- **Parameters:** $D = 10^{-14}\ \mathrm{m^2/s}$, $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$, $E_D = 73\,000\ \mathrm{cal/mol}$, $R = 1.987\ \mathrm{cal/mol \cdot K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Starting from the Arrhenius equation [Session 5 Slide 26]:

$$D = D_0 \exp\left(-\frac{E_D}{RT}\right) \implies \ln\left(\frac{D_0}{D}\right) = \frac{E_D}{RT}$$

Isolating absolute temperature T:

$$T = \frac{E_D}{R \cdot \ln(D_0/D)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Diffusivity Ratio:

$$\frac{D_0}{D} = \frac{1.55 \times 10^{-4}\ \mathrm{m^2/s}}{1.0 \times 10^{-14}\ \mathrm{m^2/s}} = 1.55 \times 10^{10}$$

2. Natural Logarithm:

$$\ln(1.55 \times 10^{10}) = \ln(1.55) + 10 \cdot \ln(10) = 0.438255 + 23.025851 = 23.464106$$

3. Absolute Temperature Calculation:

$$T = \frac{73\,000\ \mathrm{cal/mol}}{(1.987\ \mathrm{cal/mol \cdot K}) \times (23.464106)} = \frac{73\,000}{46.62318} \approx \mathbf{1565.74}\ \mathrm{K} \approx \mathbf{1566}\ \mathrm{K}$$

Conversion to Celsius:

$$T = 1565.74 - 273.15 = \mathbf{1292.59}^\circ\mathrm{C} \approx \mathbf{1293}^\circ\mathrm{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[T] = \frac{\mathrm{cal/mol}}{(\mathrm{cal/mol \cdot K})} = \mathrm{K}$. **Physical Insight:** A diffusivity of $10^{-14}\ \mathrm{m^2/s}$ needs $1293^\circ\mathrm{C}$: D depends exponentially on T, and the high activation energy ($73\ \mathrm{kcal/mol} \approx 305\ \mathrm{kJ/mol}$) means the atoms need a lot of thermal energy to jump [Slides 4, 26]. Diffusion of impurities in Si wafers is the process used to fabricate integrated circuits [Slide 29].